

MAN B&W Alpha Cylinder Lubrication System

The MAN B&W Alpha cylinder lubrication system, see Figs. 9.02.02a, 02b and 02c, is designed to supply cylinder oil intermittently, for instance every 2, 4 or 8 engine revolutions with electronically controlled timing and dosage at a defined position.

Traditional two-tank cylinder lubrication system

Separate storage and service tanks are installed for each of the different Base Number (BN) cylinder oils used onboard ships operating on both high- and low-sulphur fuels, see Fig. 9.02.02a.

The cylinder lubricating oil is pumped from the cylinder oil storage tank to the service tank, the size of which depends on the owner's and the yard's requirements, – it is normally dimensioned for about one week's cylinder lubricating oil consumption.

Oil feed to the Alpha cylinder lubrication system

Cylinder lubricating oil is fed to the Alpha cylinder lubrication system by gravity from the service tank or ACOM.

The oil fed to the injectors is pressurised by the Alpha Lubricator which is placed on the hydraulic cylinder unit (HCU) and equipped with small multi-piston pumps.

The oil pipes fitted on the engine are shown in Fig. 9.02.04.

The whole system is controlled by the Cylinder Control Unit (CCU) which controls the injection frequency based on the engine-speed signal given by the tacho signal and the fuel index.

Prior to start-up, the cylinders can be pre-lubricated and, during the running-in period, the operator can choose to increase the lubricating oil feed rate to a max. setting of 200%.

The MAN B&W Alpha Cylinder Lubricator is preferably to be controlled in accordance with the Alpha ACC (Adaptable Cylinder Oil Control) feed rate system.

The yard supply should be according to the items shown in Fig. 9.02.02a within the broken line.

Regarding the filter and the small tank for heater, please see Fig. 9.02.05.

Alpha Lubricator variants

Since the Alpha Lubricator on ME and ME-B engines are controlled by the engine control system, it is also referred to as the ME lubricator on those engines.

A more advanced version with improved injection flexibility, the Alpha Lubricator Mk 2, is being introduced on the G95/50/45/40ME-C9 and S50ME-C9 including their GI dual fuel variants.

Further information about the Alpha Lubricator Mk 2 is available in our publication:

Service Experience MAN B&W Two-stroke Engines

The publication is available at www.marine.mannes.com → 'Two-Stroke' → 'Technical Papers'.